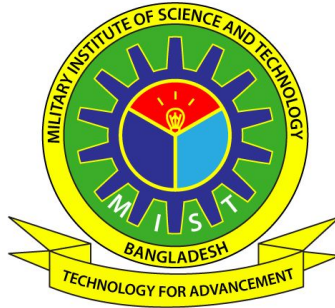


MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE-406: COMPUTER INTERFACING SESSIONAL

FINALIZED PROJECT IDEA REPORT

on

IoT Smart Energy Meter Monitoring With Theft Detection

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IoT Smart Energy Meter Monitoring with Theft Detection

1 Introduction

Managing energy and stopping waste are huge challenges in an electrical distribution system. In Bangladesh, power companies like DESCO have moved away from old mechanical meters to modern digital billing and prepaid smart meters. This has made tracking bills much cleaner and stopped manual errors. However, there is still a big catch. Even though these modern systems are digital, they generally only process and report high energy usage or system issues at the end of the month when bills are calculated. This means power theft or hidden leaks can go unnoticed for weeks, causing massive financial losses for the economy and higher costs for regular, legal customers.

To fix this, this project introduces an Internet of Things (IoT) based Smart Energy Meter with Theft Detection. Instead of waiting for a monthly report, this system tracks everything instantly. By connecting a small controller to Wi-Fi and mobile networks, it measures power use in real time, displays the cost on a local screen and sends live updates to a mobile application. A miniature version of a home setup will also be built using a real fan, light, and motor to show how the system can instantly cut off power the exact moment someone tampers with the wires or steal electricity.

1.1 Problem Statement

Even though modern power companies use digital billing, their systems work on a delay. Customers and operators only see unusual spikes or problems when the monthly data is compiled, making it impossible to stop energy waste as it happens. Additionally, because current smart meters do not track line unaccounted spikes instantly, electricity theft goes unnoticed between billing cycles. Finding illegal connections still requires workers to go out and manually check wires, which is slow and laborious. Finally, if someone tampers with the grid or defaults on a bill, operators cannot shut off that specific line remotely right away. They have to travel to the site, allowing losses to keep adding up.

1.2 Motivation

The motivation behind this project is driven by several key goals:

- Providing immediate usage information instead of monthly billing.
- Securing economic interests by reducing public revenue, national grid pressures and ensuring fair billing for legal consumers.

- Providing utility companies with instant alerts to isolate compromised lines automatically while giving regular users the precise tracking details needed to lower their energy bills and eliminate waste.
- Demonstrating a Computer Interfacing project capable of reading electrical pulses, processing signals and transfer data through an integrated network of hardware and software systems.

2 Background Study

Traditional electrical distribution systems are increasingly transitioning toward digital advanced metering infrastructures to automate transactions and track consumption patterns. In Bangladesh, distribution utilities like the Dhaka Electric Supply Company (DESCO) have widely deployed digital prepaid meters to optimize revenue collection and minimize manual billing overheads. However, because contemporary prepayment frameworks are primarily designed for financial auditing rather than active security surveillance, the vulnerability remains where theft is unnoticed between billings [12]

To solve this on a smaller scale, student and maker communities often build simple digital wattmeters using an Arduino board and a current sensor to measure power locally [2, 3]. These basic projects are incredibly cheap and easy to make, but they have no way to send data over the internet. A person must stand in front of the box to see the numbers, and these devices cannot detect or stop theft.

To send alerts over long distances without needing a local internet router, builders often use mobile phone components. Popular online projects show how to connect an Arduino to a SIM800L cell module to text billing updates to a user [4], and research papers explore sending emergency text warnings over cellular networks [5]. Notably, it works anywhere there is cell service, even in remote areas. However, text messages only react after an anomaly happens. They cannot show a continuous, live stream of data or graphs, and sending frequent emergency texts can get expensive over time.

On the other hand, many developers use inexpensive Wi-Fi chips to send data directly to the cloud. Online tutorials explain how to use an ESP8266 Wi-Fi module to stream live electricity data straight to open IoT platforms like ThingSpeak [6], while advanced papers look at feeding this data into organized web dashboards [7]. This is highly useful because users can open a phone app from anywhere in the world to see live charts of their energy consumption and costs. The downside is that it is entirely dependent on home Wi-Fi. If the router loses power or the internet drops, all remote tracking stops immediately.

For hardware protection, builders use physical sensors to detect tampering. Simple setups place a hidden mechanical switch inside the meter box that triggers an alarm if the cover is opened [8]. More advanced research looks at comparing the current entering a line versus the current leaving it to spot illegal taps [9]. While these switches are incredibly sensitive and sound a loud buzzer instantly, they are usually blind and isolated. They can make noise locally, but they cannot notify a remote operator or take automated action to protect the rest of the grid.

Finally, many basic IoT projects focus on using electronic switches, called relays, to turn appliances on and off. Simple tutorials use these to protect against overloads [10], and academic papers show how they can automatically turn off certain rooms to save power during peak hours [11]. These setups are safe for testing dummy fans, lights, and motors with a small controller. However, they are usually controlled by basic timers or manual buttons. They lack the smart processing loops needed to act as an automatic circuit breaker, meaning they cannot flip at the exact moment a theft is detected.

3 Objectives

The main goal of this project is to create a responsive and automated energy network that addresses the gaps in modern utility reporting. The specific objectives are:

1. To eliminate billing delays by achieving immediate real time tracking of household electricity use.
2. To keep both utility companies and consumers informed instantly by sending automated updates the moment power issues or line mismatches occur.
3. To prevent financial and resource losses by automatically cutting off specific lines when unauthorized tampering or theft is found.
4. To prove the reliability of this automated safety logic through a safe physical simulation using functional home appliances.

4 Proposed Methodology

The complete system is designed to seamlessly manage data from initial physical inputs to final localized and cloud-based outputs. The operational process begins at the input layer, where an electricity meter measures power consumption and generates corresponding electrical pulses. These high-voltage pulses are securely isolated and converted into low-voltage digital signals using an optocoupler circuit before being passed directly to the digital input pins of the Arduino UNO microcontroller. Simultaneously, the microcontroller monitors a cost-setting button array and a structural theft detection switch to identify any

unauthorized case tampering.

Once the incoming pulse data is received, the Arduino UNO executes internal embedded C programs to calculate total power consumption in kilowatt-hours (kWh) alongside the cumulative financial cost. These metrics are immediately formatted and sent over three distinct pathways. Locally, the real-time data is output to a 16x2 LCD screen using an I2C adapter to minimize pin usage. Globally, the metrics are pushed over a serial interface to an ESP8266 Wi-Fi module, which continuously updates a secure cloud-based IoT web dashboard.

The system constantly evaluates a built-in software conditional statement to determine if the theft switch has been triggered. Under normal monitoring conditions, the program loops back to read subsequent energy pulses while maintaining continuous data feeds to the web dashboard. If the theft detection switch is flipped, the program branches into an emergency routine that sets off a local 5V buzzer alarm and routes an urgent command to the onboard GSM module to broadcast an alert SMS to the user's phone. Concurrently, the Arduino UNO sends a high digital signal to a 4-channel relay module, acting as an automated circuit breaker to cut off power to the dummy appliances, including the fans, lights, and motors. System restoration or safe shutdowns can then be executed remotely by sending an official control command back through the internet dashboard to the ESP8266 receiver module.

4.1 Flow Diagram

The operational pipeline of the system is modeled after the structured sequence found in Fig. 1. When the primary power supply is turned on, the system initializes its input and output peripherals, allowing the energy meter to begin capturing electrical consumption. The Arduino UNO reads these incoming data pulses, computes total units alongside power pricing, and updates the local LCD display. This data is systematically uploaded to the remote IoT dashboard using the ESP8266 wireless link.

The microcontroller then goes through a conditional decision step to determine if power theft has occurred. If no theft is detected, the workflow routes back to normal monitoring to continue reading consumption pulses in a regular loop. If power theft is validated, the system breaks the loop and activates the local buzzer alarm. It then sends an emergency SMS alert via the GSM module, updates the theft status on the IoT web interface, and disengages the relay module to cut off the consumer's load grid. The process concludes once the end-user receives the notifications and secures the physical hardware infrastructure.

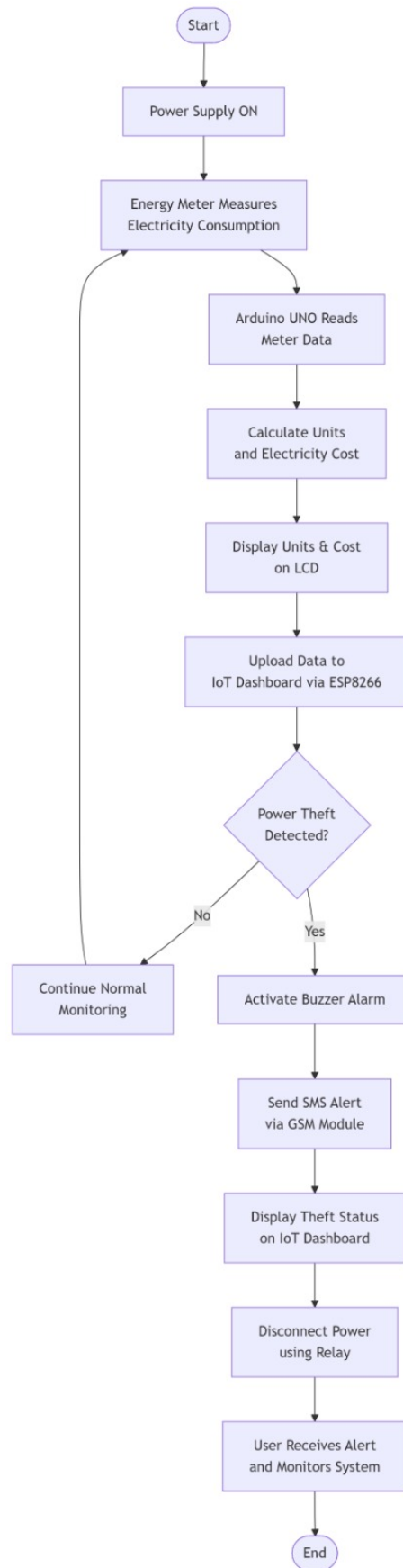


Figure 1: Flow diagram of the project

4.2 System Architecture

The interconnected hardware and software layers follow the exact layout detailed in Fig. 2. The input layer contains the cost-setting buttons, the hardware theft switch, and the primary energy meter, which collectively feed physical information into the processing layer. At the center of the processing layer is the Arduino UNO, which handles mathematical calculations, keeps track of timing, and coordinates peripheral device actions.

The processing layer interacts directly with the communication block, which houses the ESP8266 Wi-Fi module and the GSM module. The ESP8266 handles two-way communication, sending local status metrics up to the Cloud Platform IoT Web Dashboard and receiving remote ON/OFF commands from the user. At the same time, the GSM module provides a direct cellular connection to send SMS notifications to the user's mobile device when the system detects tampering. Finally, the output layer features an LCD screen for on-site monitoring, a buzzer for audio alerts, and a multi-channel relay module that connects directly to the dummy loads, simulating real household lights, fans, and motors.

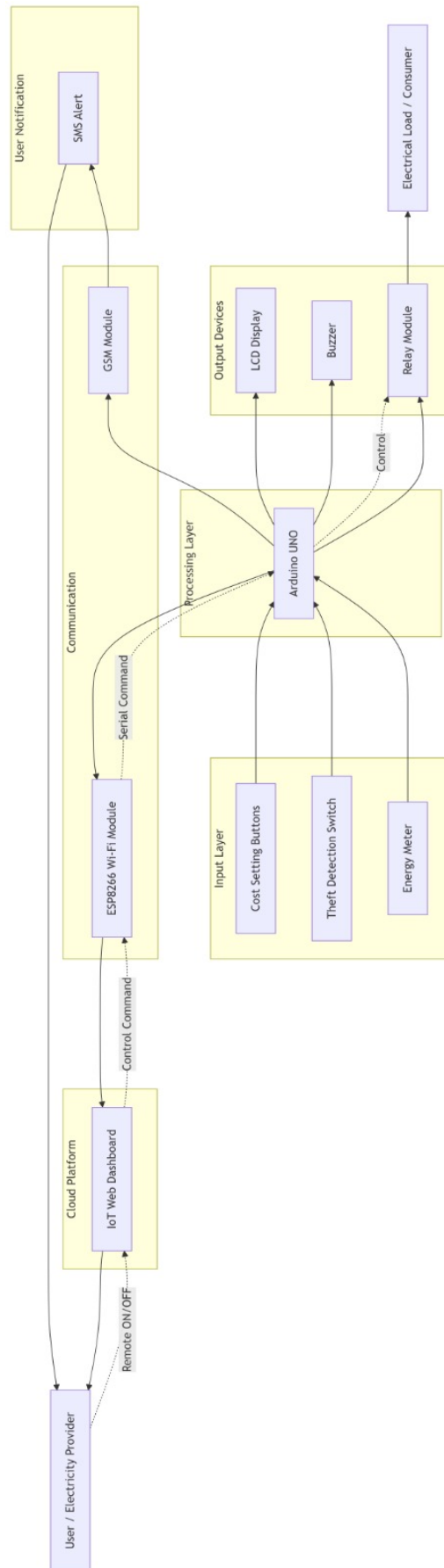


Figure 2: System architecture of the project

5 Component Description

The Arduino UNO R3 acts as the main processing unit for the entire project. It is equipped with an ATmega328P microcontroller, 14 digital input/output pins, and 6 analog inputs, operating at a stable clock speed of 16 MHz. It reads input pulses from the isolation circuitry, calculates accurate billing data, and controls the communication and relay peripherals.

The ESP8266 Wi-Fi Module enables internet connectivity for the hardware grid. Operating at 2.4 GHz with an onboard integrated TCP/IP protocol stack, it communicates with the Arduino UNO via serial commands to upload real-time consumption and theft logs to the internet web dashboard.

The SIM800L GSM Module provides cellular connectivity for sending emergency text alerts. It operates on standard quad-band mobile frequencies and links to the microcontroller using a serial communication interface to send instant notifications to the user's phone if line tampering is detected.

The PC817 Optocoupler is an optoelectronic isolation integrated circuit that contains a light-emitting diode optically coupled to a phototransistor. This component is necessary to safely link the high-voltage energy meter to the low-voltage Arduino pins, converting meter pulses into safe digital inputs without risk of electrical damage.

The 16x2 LCD with an integrated I2C Module serves as the main on-site display. The I2C adapter reduces the connection footprint to just two data lines, saving valuable pins on the microcontroller for other dummy components.

The 4-Channel 5V Relay Module functions as an automated multi-line circuit breaker. Each channel contains an electromagnetic switch driven by a 5V input signal, allowing the low-power Arduino to safely connect or disconnect power to individual high-power dummy appliances.

The Dummy Load Components consist of a 5V/12V DC brushless cooling fan, a small DC toy motor, and an array of high-brightness LEDs. These components form our dummy system, standing in for household cooling units, industrial motor loads, and lighting grids to clearly show how the system manages power and handles load shedding during a simulated theft event.

The Arduino IDE Studio Compiler is the core software platform used for development. This environment is used to write, debug, and upload the embedded C control code to the Arduino UNO, managing all data processing, timing loops, and communication protocols.

6 Budget Analysis

The budget allocation below outlines the cost estimates for all required hardware components based on current market rates on RoboticsBD [1].

Table 1: Estimated Budget Analysis

S/N	Component Name	Specification/Model	Qty	Unit Price (BDT)	Total (BDT)
1	Microcontroller Board	Arduino UNO R3 (Clone)	1	650	650
2	Wireless Wi-Fi Module	ESP8266 ESP-01	1	220	220
3	Cellular GSM Module	SIM800L Micro GSM	1	550	550
4	Optical Isolator IC	PC817 Optocoupler	2	15	30
5	Character LCD Assembly	16x2 LCD with I2C Adapter	1	350	350
6	System Switching Unit	4-Channel 5V Relay Module	1	280	280
7	Simulation Cooling Fan	5V DC Brushless Fan	1	90	90
8	Simulation Load Motor	5V DC Toy Motor	1	60	60
9	Simulation Grid Light	5V Cob LED Module Strip	2	45	90
10	Setup Input Matrix	Tactile Push Buttons & Toggle Switches	5	12	60
11	Acoustic Alarm Device	5V Active Buzzer	1	30	30
12	Prototype Breadboard	Full-Size 830-Point Breadboard	1	140	140
13	Connecting Wire Pack	Male-to-Male & Male-to-Female Jumpers	1	120	120
Total Estimated Hardware Cost					2,670

This proposed project is highly practical and feasible. The total projected cost is only BDT 2,670 because it relies on standard, easily sourced components available on RoboticsBD. Since all development tools

and software are open source, there are no additional software services expenses, ensuring the project remains within budget limits.

7 Workplan

This project follows a 13-week plan, opening with five weeks of research of ideas and component sourcing. Weeks 6 through 9 handle the core implementation by overlapping physical hardware integration with local embedded C programming and cloud dashboard setup. The final month focuses entirely on system validation, testing the automated relay cutoff logic against the dummy appliances (fans, lights, and motors), before ending with system calibration and the final project presentation.

Table 2: Tentative Gantt chart for the project

Sl. No.	Activity Name	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
1	Idea Selection & Background Study	✓	✓	✓	✓									
2	Component Sourcing					✓								
3	Hardware Integration & Layout						✓	✓						
4	Local Embedded Code Design							✓	✓					
5	Cloud Platform & IoT Setup								✓	✓				
6	Dummy Load System Testing										✓	✓		
7	System Calibration & Tuning												✓	
8	Final Report & Presentation													✓

8 Conclusion

This proposal presents an IoT Smart Energy Meter with Theft Detection designed to eliminate the delayed reporting gaps found in modern digital utility grids. The system operates by reading physical electricity consumption pulses through an isolation circuit, processing the metrics on an ASM328P based Arduino UNO microcontroller, and using Wi-Fi and cellular modules to update a mobile application dashboard

and a local display. The expected output is a fully automated safety network that provides instantaneous tracking data, sounds a buzzer, sends an emergency mobile text message, and triggers a relay switch to isolate specific power lines the moment tampering is detected. This project is highly useful because it protects public revenue, reduces grid distribution losses, and gives consumers the live tracking information needed to regulate their energy budgets. Additionally, it serves as a practical demonstration of computer interfacing concepts by showing how a central processor can measure real world physical events, manage external appliances, and transfer data through a network of integrated hardware and software. Future improvements could enhance this design by introducing intelligent Machine Learning algorithms to learn and predict energy needs, enabling backup power, or allowing wireless software updates on the controller.

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